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STSO-240: Social Consequences of Technology

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Research Paper

04/27/2026

The Influence of Social Media Algorithms on User Behavior and Political Perspectives

Introduction to the Technology

Social media has become one of the most prominent technological developments of the twenty-first century, as it has entirely changed how people communicate, consume information, and form their views of the world surrounding them. Popular platforms like Instagram, TikTok, Facebook, and X have started to become primary news and information sources for a large portion of the global population, especially for younger generations. However, at the heart of these platforms, there is a set of sophisticated recommendation and content-curation algorithms put in place to determine what each of their users see in their personalized feeds at any given moment. These algorithms are said to be designed to keep users engaged, but the ways they have been able to shape what people are exposed to have had far-reaching consequences that go well beyond the purpose of simple entertainment.

At their core, these social media algorithms function as “automated gatekeepers,” as they constantly sort and rank content for users based on a mix of different factors: how long a user looks at a post, who they follow, what they like or share, and even the patterns or behaviors of users with similar profiles. The use of these algorithms does result in a highly curated, personal digital experience for the user, but it also means that two people who are using the same platform

can end up in completely different ecosystems of information. This paper will explore how these algorithmic systems are shaping user behavior, influencing political perspectives, and shifting civic engagement.

The Issue the Technology "Solves"

The initial reason that social media algorithms were created was to solve the problem of overloading their users with information. As the internet kept growing and content became available at a much larger scale, social media companies wanted to make a way for users to find content that would specifically be more interesting to them. Without some form of algorithmic content curation, social media feeds would have unusable floods of random posts from every person or page that the specific user follows. Algorithms were introduced to help prevent this by filtering and ranking each user's content so the most engaging is put to the top.

From the perspective of user experience, this does actually make a lot of sense. Not many users will want to scroll through a lot of uninteresting posts until they find something interesting or relevant to them. As Poleac and Ghergut-Babii mention, social media platforms have assumed control over lots of online interaction, which is significantly shaping how users are making their decisions, consuming their information, and engaging with others in digital spaces. Their goal was to make their platforms have more of a pull and be more useful so people would have a reason to keep coming back. Algorithms were able to accomplish this by learning each user's preferences and curating content that felt more tailored to them.

However, this so-called "solution" created its own list of larger, more complicated issues. These algorithms optimizing for the user's engagement does not necessarily mean they're optimizing for information accuracy, diversity, or the health of democracy. Political, emotionally charged, and divisive content usually generates more user engagement than more neutral,

unopinionated content, and the algorithms were able to pick up on this pretty quickly. What started as a way to filter out content that would be irrelevant to users became a mechanism that can amplify extreme viewpoints, filtering out exposure to any other political opinions that a healthy democracy should have.

The Science Behind the Technology

Understanding how social media algorithms work requires looking into the code and data infrastructure that actually runs them. At their core, these systems are operated by machine learning models, more specifically collaborative filtering and reinforcement learning systems, and are trained on ridiculously large datasets of user behaviors. These systems track each click, pause, scroll, and interaction, where they then get fed back into the model, helping it get better at predicting what a specific user will engage with next. The more data that the system can collect on a user, the more precisely the system can target content to the users preferences.

In a research study conducted by Thorson, Cotter, Medeiros and Pak, they examined how Facebook specifically builds out what they refer to as “interest inferences” about their users. These are essentially algorithmic labels that categorize users by their topic interests, which includes their political interests. Their research found that users who were algorithmically categorized as being interested in news or politics were significantly more likely to have political content curated into their feeds, above what their own self-reported interest would suspect. This reveals how the algorithm is not just responding to what users say they want, but is actively classifying and sorting the user in a way that shapes their information environment, a lot of the time without them realizing it.

Another important phenomenon, or more likely a mechanism, relevant to this topic is what researchers have called the “echo chamber” effect. Researchers Cinelli, De Francisci

Morales, Galeazzi, Quattrociocchi, and Starnini conducted a large-scale analysis of over one hundred million pieces of content across popular social media platforms including Facebook, Twitter, Reddit, and Gab. Through this, they found that the way algorithms and social interaction patterns tend to work together commonly tends to push users toward each other in clusters of like-minded people. Because of this, users end up in what are essentially informational bubbles, where their existing beliefs end up getting reinforced over and over through repeated exposure to content from sources or people who already agree with them. This study also found that there were significantly high levels of this kind of "homophily" on Facebook.

Political microtargeting represents another dimension of the influence of algorithmic content curation used across social media. In research conducted by Zarouali, Dobber, De Pauw, and de Vreese, they decided to investigate how personality-profiling algorithms can be implemented to match political advertisements to specific personality traits of individual users. They found in their research that users are more strongly persuaded by political ads that are tailored to the profile of their personality, inferring significant implications for political campaigns done on social media. The technology used accomplishes this by running users' text and behavioral data through different personality inference models (similar to the OCEAN model in the Cambridge Analytica case) to predict traits like openness and extraversion, and then serving them political content curated specifically to resonate with those traits.

The Social Consequences of this Technology

The social consequences of algorithm-driven social media are easily the most important aspect of this entire issue. Due to how recently significant and complex this issue is, researchers are still examining different effects of algorithmic content curation. However, there are still some found effects that have been documented across various studies.

One of the most commonly talked about consequences of this technology is political polarization, the idea that social media algorithms push their users further toward the extreme ends of the political spectrum by curating content that they already believe, while cutting their exposure off to opposing viewpoints. In an experimental study done by Cho, Ahmed, Hilbert, Liu, and Luu, they conducted research using YouTube's actual recommendation algorithm that they use to recommend videos to their users. In this study, they found that exposure to algorithmically recommended content reinforced and polarized political opinions among their participants. The mechanism used here is pretty easy to understand: if you watch a few videos leaning toward one political direction, the algorithm automatically starts feeding you more and more, usually extreme content in that same direction, as extreme content tends to generate more engagement. This effect of algorithmic curation shows how it can shift users' political "normal" over time.

However, there was also research that countered some of the concerns throughout this paper. A study by Feezell, Wagner, and Conroy focused on algorithmically driven news sources, distinguishing between "socially driven" algorithms, which would be content recommended based on what your social network engages with, and user-driven algorithms, which would be content recommended based on your own history. These researchers found something a little surprising, which is that rather than contributing to partisan polarization, getting news from algorithm-driven sources was actually associated with higher levels of political participation. They also concluded that neither type of algorithmic news delivery significantly increased partisan polarization, which would slightly counter the argument that algorithms being used this way will destroy democracy. Research done by Nguyen and Vu also challenged some of the issues brought up, more specifically about the "echo chamber" effect. Through a study where

they examined Eurobarometer survey data, they were not able to find much strong evidence that relying on social media as a primary news source necessarily leads to more political polarization than relying on traditional media. They argue that much of the popular arguments around echo chambers has been oversimplified, claiming that a lot of research has actually found that social media can expose people to a broader range of views than the consumption of self-selected media. This is quite a significant counterpoint to the more "mainstream" narrative, however, this research also does not erase all of the other documented concerns.

Where I believe the consequences become more serious is in the aspects of political microtargeting and behavior manipulation. In a research study led by Zarouali, their findings showed clearly that algorithms can be used to craft political messages that bypass rational deliberation and are able to target people at the level of their personality traits and emotional vulnerabilities. This would have a significantly different level of political influence than traditional advertising would. It would no longer just be about which message you see, but about receiving a version of a message that's specifically engineered to be maximally persuasive for you, all based on data that you did not knowingly provide for this purpose. The Cambridge Analytica scandal was a real world example of this risk, as it demonstrated how these techniques are already being used at scale in major democratic elections.

Another huge social consequence of this technology is its effects on civic participation in a broad sense. A study by Jung, Dai, and Albarracín reviewed research regarding how social media algorithms shape offline civic behavior, with things like voting, donating to causes, and showing up to protests. In this study, they found that while social media has created significant new pathways for civic engagement, algorithmic curation has been playing a complicated role in determining whether online activism actually translates into real-world action. The algorithm's

tendency to reward emotionally driven, easily consumable content can often encourage what's known as "slacktivism," which is more just liking a post or sharing a video, rather than committing to a deeper form of civic engagement. On the other hand, the research also found that these same algorithms can build "social capital" and foster generalized action goals when thoughtfully designed.

An issue that is tightly connected to all of the other consequences is the spread of misinformation across social media. Because of how algorithms prioritize user engagement, as well as how emotionally provocative and outrage-inducing content tends to generate more engagement than dryer, factual reporting, misleading information gets a structural advantage across algorithmically curated environments. The echo chamber effect then pairs with this, as once misinformation enters an information cluster, the algorithm begins to keep circulating it within that group, rarely introducing any kind of corrective information that might come from outside sources. This creates opportunities for false beliefs to spread and become more believed in, which could lead to many issues regarding public health, elections, social trust, and more.

The influence of these algorithms does not get evenly distributed either. With the way that these systems work, people who are already politically engaged tend to get more political content curated towards them, while people who are less engaged may get less exposure to political information altogether. This can lead to many implications for anyone who participates in democratic processes and whose voices get amplified. In the earlier study led by Thorson, he found that the categorization of algorithms shapes political content exposure independently from self-reported interest means that platforms are actively making decisions about who counts as a political person, which could have real democratic stakes.

After conducting my research, my overall take is that algorithms used in social media are not inherently destructive, but are being used in ways that prioritize corporate profit over social well-being, which can and has led to measurable harms. The use of machine learning to personalize content isn't the problem, but having the optimization target be engagement instead of accuracy is. When a company builds a system designed to maximize the time users spend on their platform, people should not be surprised when that system ends up amplifying outrage, division, and misinformation, all things that are known for capturing human attention.

My biggest concern from an STS perspective is the lack of transparency or accountability. Most users have no idea how they are being classified, what content gets withheld from them, or how their data gets used against them to engineer their beliefs. The personality-targeted political advertising documented by Zarouali showed how it operates mostly in the dark, meaning users have almost no way to opt out of what is happening, taking away from their consent and autonomy. When technology begins shaping democratic participation at this level without public understanding or regulatory oversight, I believe there needs to be an obligation to push for greater accountability of the corporation operating it.

I don't believe the right answer is to "abolish" these algorithms, considering that's not very realistic, but I do believe that these platforms need to be designed for better democratic health instead of just engagement, whether it's diversifying content feeds by design, labeling algorithmically promoted content, giving users more control over their recommendation settings, or even transparency reports about how political content is curated. Regardless, giving users better tools and information to make their own choices seems like a better direction for social media to take. Ultimately, the social consequences of social media algorithms are a design choice, not a technological inevitability, meaning they can be designed for the benefit of users.

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